AI & DATA SCIENCE • COMPUTER VISION PROJECT

Deep Learning-Based Crowd Density Estimation and People Counting Using Computer Vision

An End-to-End Computer Vision Framework for Automated Crowd Safety Monitoring

PRESENTED BY

B. Praneeth Reddy

ROLL NO.	SECTION	SUBJECT	DEPARTMENT
160124771028	I1	AAN	AI & DS

Introduction

ρ

Crowd Density Estimation

Measures how closely people are packed within a monitored scene, expressed as a resolution-independent index rather than a raw headcount.

N

People Counting

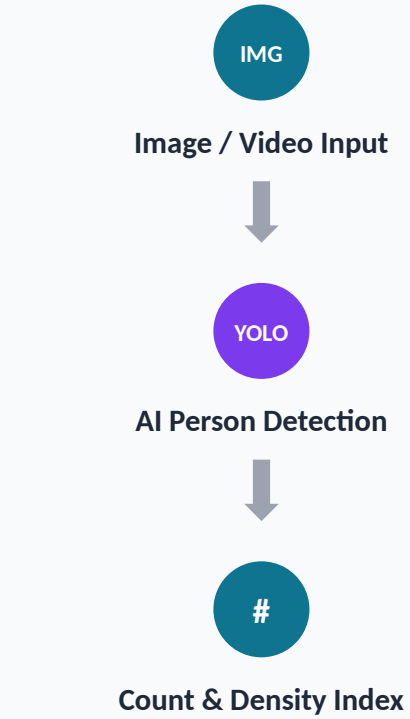
Automatically detects and counts the number of individuals present in an uploaded image or video frame, without manual tallying.

AI

Why Computer Vision + Deep Learning

A pretrained neural network turns ordinary camera footage into continuous, objective, round-the-clock crowd intelligence — something manual observation cannot sustain.

FROM FOOTAGE TO INSIGHT



Problem Statement & Motivation

! The Problem

- Human operators lose visual focus after prolonged CCTV monitoring, so bottlenecks are often missed until movement has already stalled.
- Turnstiles and manual counters record entry/exit totals only — they reveal nothing about internal spatial clustering.
- Manual intervention typically begins only after panic or injury has already started.
- Raw headcount alone is misleading: 50 people in an open square is safe; 50 people in a narrow stairwell is dangerous.

✓ The Motivation

- Mass gatherings at transit hubs, pilgrimage centres, airports, stadiums and public squares are prone to sudden density surges.
- An automated system gives security teams an early, objective warning before conditions turn dangerous.
- Non-intrusive: works with existing RGB camera infrastructure — no RFID tags or mobile tracking required.
- Runs fully offline for data privacy, with adaptive CPU/GPU execution so it stays accessible on standard hardware.

Objectives

1

Detect People

Localize pedestrians in a scene using a pretrained YOLO26 model.

2

Count People

Produce instantaneous and cumulative unique counts via tracking.

3

Estimate Crowd Density

Compute a resolution-normalized density index and surface occupancy.

4

Generate Heatmaps

Render 2D Gaussian spatial density heatmaps over the scene.

5

Track People

Maintain persistent multi-person IDs across video frames.

6

Classify Crowd Risk

Categorize scenes into LOW / MODERATE / HIGH / CRITICAL tiers.

7

Process Images & Video

Deliver it all through an interactive Streamlit web application.

System Architecture & Methodology

Stage	Technique Used	Output Produced
1. Input	Accept uploaded image or video file	*Raw frame(s) in memory*
2. Validation	Check format, integrity & decode the file	*Verified, readable frame*
3. Person Detection	YOLO26 inference, confidence ≥ 0.35	*Raw person detections*
4. Bounding Boxes	NMS filtering removes duplicates (IoU ≥ 0.45)	*Clean boxes [x1,y1,x2,y2] + confidence*
5. Counting	Tally detections in the current frame	*Instantaneous person count (N)*
6. Density Calculation	$\rho = (N \times 10{,}000) / (W \times H)$	*Resolution-normalized density index*
7. Occupancy Calculation	Binary mask over the union of all boxes	*True surface occupancy Ω (%)*
8. Spatial Distribution	3×3 grid split + coefficient of variation	*Hot-zones & dispersion type*
9. Tracking	Greedy IoU + centroid matching (IoU ≥ 0.30)	*Persistent person IDs, no double-count*
10. Risk Classification	Multi-criteria check on N, ρ and Ω	*LOW / MODERATE / HIGH / CRITICAL*
11. Visualization	Gaussian heatmap + HUD overlay rendering	*Annotated media & live dashboard*

05 · TECH STACK

Technologies & Tools

Py

Python

Core programming language

YL

Ultralytics YOLO26

Pretrained person detection

Pt

PyTorch

Deep learning backbone

CV

OpenCV

Image & video processing

St

Streamlit

Interactive web application

Np

NumPy

Numerical computing

Pd

Pandas

Tabular data handling

Mp

Matplotlib

Charts & metric plots

Pl

Pillow (PIL)

Image I/O utilities

Sc

SciPy

Scientific computing

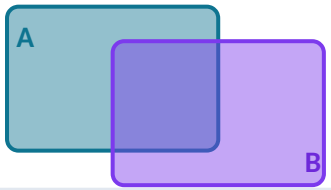
YOLO Person Detection & Tracking

YLO Detection — YOLO26

- Single-stage deep CNN: locates every person in one forward pass over the frame.
- Filters strictly for COCO Class 0 (“person”) via classes=[PERSON_CLASS_ID].
- Confidence threshold: default 0.35, adjustable between 0.10 – 0.95.
- Outputs per-person bounding boxes [x1, y1, x2, y2] with confidence scores.

ID Tracking — IoU + Centroid

- Greedy bipartite matching links each detection to the closest existing track.
- IoU ≥ 0.30 required to associate a detection with an existing track.
- Persistent IDs (Person 1, Person 2 …) assigned on first appearance.
- Tracks survive up to 15 missed frames before being pruned — preventing double-counting.



Intersection-over-Union: IoU = Area(A ∩ B) / Area(A ∪ B) — the overlap ratio used both to suppress duplicate detections (NMS) and to match people across consecutive frames.

07 · CORE ANALYTICS

Density, Heatmap & Risk Classification

Density Index ρ

`ρ = (N × 10,000) / (W × H)`

People per 100×100 px window — resolution-invariant.

Surface Occupancy Ω

`Ω = (Occupied / W×H) ×100%`

Binary-mask coverage; overlaps not double counted.

Spatial Distribution

`3×3 grid · CV = σ / μ`

Flags hot-zones; classifies Uniform vs Clustered.

Gaussian Heatmap D(x,y)

`D = Σ exp(−dist² / 2σ²)`

Per-person Gaussian, blurred & color-mapped (JET).

MULTI-TIER OPERATIONAL RISK CLASSIFICATION — thresholds defined in config.py

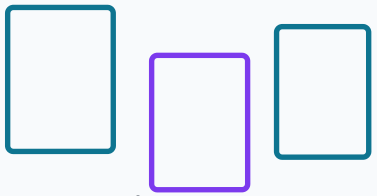
Crowd Level	Person Count	Density Index ρ	Occupancy Ω	Operational Meaning
LOW	≤ 9	≤ 0.035	≤ 8.0%	Sparse crowd; free pedestrian movement.
MODERATE	10 – 24	≤ 0.085	≤ 18.0%	Normal density; fluid movement maintained.
HIGH	25 – 49	≤ 0.160	≤ 32.0%	Restricted flow; bottleneck warnings.
CRITICAL	50+	> 0.160	> 32.0%	Overcrowded; high crush hazard — intervention required.

Results & Output



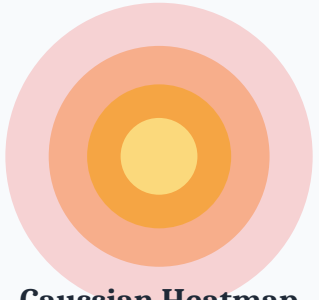
Original Frame

Validated input image/video



Detections + Boxes

YOLO boxes, confidence, IDs



Gaussian Heatmap

Spatial density hot-zones



Blended Overlay

Heatmap + detections combined

Schematic of the 2×2 composite panel generated by visualization.py

ADDITIONAL OUTPUT ARTIFACTS

- *_annotated.jpg — boxes, confidence tags, HUD badge
- *_heatmap.jpg — 2D Gaussian spatial heatmap
- *_combined.jpg — 2×2 presentation composite
- *_analysis.json — quantitative analytics metadata
- processed_video_*.mp4 — annotated video stream
- crowd_metrics_plot_*.png — count & density vs time
- video_summary_*.json — aggregate temporal report

CROWD RISK BADGES

LOW | MODERATE | HIGH | CRITICAL

STREAMLIT DASHBOARD TABS

Home | Image Analysis | Video Analysis | Dashboard | Technical Docs

Conclusion & Future Scope

✓ What We Achieved

- A complete offline CV pipeline combining detection, density metrics, heatmaps and tracking.
- Packaged into an interactive Streamlit app with image, video and dashboard views.
- Validated with an automated test suite across hardware, edge cases and video.

± Advantages & Limits

- 100% offline with adaptive CUDA / CPU execution and resolution-invariant metrics.
- Perspective distortion: nearer people appear larger than farther ones.
- Heavy occlusion in extremely dense crowds can reduce accuracy.

→ Future Scope

- Ground-plane homography for top-down perspective correction.
- Hybrid density-regression models (e.g. CSRNet) for ultra-dense aerial views.
- Edge deployment via TensorRT / ONNX on devices like NVIDIA Jetson.

Thank You

We are now open for your questions